

Amazon Sales Analysis Case Study

Comprehensive Business Analysis Report

Prepared By: Abhijeet Kedar

Role: Business Analyst Case Study

Tools Used: Microsoft Excel, Pivot Tables, Data Analysis, Statistical Analysis, Data Visualization

Analysis Period: 01 May 2019 – 31 August 2019

Executive Summary

This project analyzes Amazon sales and customer engagement data between May 2019 and August 2019. The objective was to identify high-performing products, evaluate sales trends, assess the impact of promotional events, investigate category growth patterns, identify data quality issues, and determine whether pricing influences customer conversion behavior.

The analysis was conducted using Sales and Glance datasets containing over 81,000 combined records. Multiple analytical techniques including pivot table analysis, trend analysis, growth calculations, conversion analysis, and Pearson Correlation were used to derive business insights.

Key findings include the identification of a major sales event during mid-July, the discovery of underperforming product subcategories, several data quality concerns, and evidence that pricing had minimal impact on customer conversion for the analyzed SKU.

The findings provide actionable recommendations that can help improve product portfolio management, data governance, and future promotional strategies.

Project Overview

Business Objective

The primary objective of this case study was to analyze Amazon sales performance and customer traffic data to answer specific business questions regarding product performance, promotional effectiveness, category growth, and customer purchasing behavior.

The analysis aimed to transform raw transactional and glance data into meaningful business insights that could support strategic decision-making.

Scope of Analysis

The project focused on:

- Product-level performance analysis
- Revenue and sales trend evaluation
- Promotional event identification
- Category and subcategory growth assessment
- Data quality validation
- Pricing and conversion relationship analysis

Dataset Overview

Two datasets were provided for analysis:

1. Sales Dataset

The Sales dataset contains information related to product sales, revenue generation, and units sold.

Key Metrics

Metric	Value
Total Sales Records	41,115
Analysis Start Date	01-May-2019
Analysis End Date	31-Aug-2019
Total SKUs	465
Total Categories	10
Total Subcategories	24

2. Glance Dataset

The Glance dataset contains customer engagement metrics including product views and interactions.

Key Metrics

Metric	Value
Total Glance Records	40,745

Methodology

The following methodology was used throughout the project:

Data Preparation

- Validated record counts
- Verified date ranges
- Reviewed category and subcategory structures
- Created calculated fields where required

Analytical Techniques

- Pivot Table Analysis
- Revenue Trend Analysis
- Growth Rate Calculation
- Unit Conversion Analysis
- Average Selling Price (ASP) Analysis
- Pearson Correlation Analysis

Key Formula Used

Average Selling Price (ASP) - $ASP = \text{Ordered Revenue} / \text{Ordered Units}$

Unit Conversion - $\text{Conversion Rate} = \text{Units} / \text{Views}$

Growth Percentage - $\text{Growth \%} = (\text{August Revenue} - \text{May Revenue}) / \text{May Revenue}$

1. Identify the Most Expensive SKU

Objective

Determine the SKU with the highest Average Selling Price (ASP).

Approach

A SKU-level summary was created using pivot table analysis. Revenue and units sold were aggregated for each SKU, and ASP was calculated.

Findings

Highest ASP SKU

Metric	Value
SKU	C03CBL[721
Revenue	575,044.59
Units Sold	501

Business Insight

The identified SKU generated substantial revenue despite relatively low sales volume. This suggests that premium-priced products can contribute significantly to overall business performance.

Conclusion

SKU C03CBL[721 was identified as the most expensive SKU in the portfolio and represents a strong premium-product revenue contributor.

2. Revenue Generating and Stopped Selling SKUs

Objective

Identify revenue-generating SKUs and products that stopped selling during the analysis period.

Approach

SKU performance was analyzed using monthly sales data. A SKU was considered stopped selling if:

July Units > 0

August Units = 0

Findings

Metric	Value
Revenue Generating SKUs	365
Total SKUs	465
Revenue Generating Percentage	78.49%

Business Insight

Most products successfully generated revenue during the analysis period. However, several products stopped selling after July, indicating possible inventory shortages, product discontinuation, or reduced demand.

Conclusion

The majority of SKUs contributed to revenue generation, while a subset of products requires investigation due to discontinued sales activity.

3. Identification of Sale Event

Objective

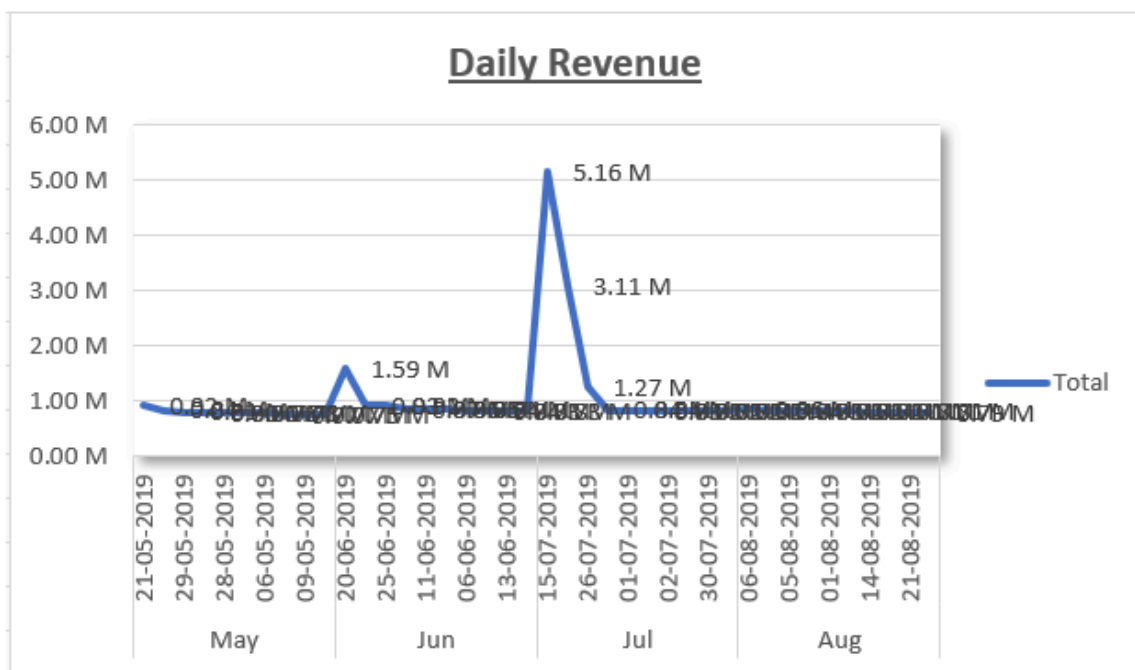
Identify dates during which a major sale event occurred.

Approach

Daily revenue trends were analyzed using pivot tables and line charts.

Findings

Daily Revenue Trend Chart



These dates exhibited Significant revenue spikes, higher revenue than surrounding days:

- 15 July 2019
- 16 July 2019

Business Insight

The abnormal increase in revenue strongly suggests a promotional or sale event that successfully increased customer purchases.

Conclusion

A major sales event occurred during mid-July and generated a noticeable increase in revenue.

4. Impact of Sale Event on Future Sales

Objective

Determine whether the sale event negatively impacted future sales performance.

Approach

Revenue levels before and after the sale event were compared using daily revenue analysis.

Findings

Revenue Before vs After Sale Event chart



Revenue declined after the event; however, the decline appeared to be a return to normal levels rather than a sustained reduction.

Business Insight

The promotional event successfully generated short-term demand without causing severe sales cannibalization.

Conclusion

There is no strong evidence that the sale event significantly harmed future sales performance.

5. Slowest Growing Subcategory Analysis

Objective

Identify the slowest-growing subcategory relative to its parent category.

Approach

Category and subcategory growth percentages were calculated and compared.

Findings

Most Concerning Subcategory

Metric	Value
Category	Computer Peripherals
Subcategory	Keyboards - DELETED
Category Growth	64.30%
Subcategory Growth	-99.20%

Business Insight

While the overall category experienced strong growth, this subcategory experienced a significant decline. This discrepancy indicates a potential issue that requires further investigation.

Possible reasons include:

- Product discontinuation
- Inventory constraints
- Reduced customer demand
- Product classification issues

Conclusion

Keyboards - DELETED represents the most concerning subcategory within the portfolio.

6. Data Quality Issues and Anomalies

Objective

Identify data quality concerns that could impact analytical accuracy.

Findings

Issue 1: Zero Unit Records

Some records contained zero ordered units, which can cause calculation errors when deriving ASP.

Issue 2: Discontinued Product Classifications

The presence of "DELETED" within subcategory names suggests incomplete data cleanup processes.

Issue 3: Missing Glance Records

The difference between sales and glance record counts indicates potential gaps in customer traffic tracking.

Issue 4: Extreme Growth Values

Some subcategories displayed unusually high growth percentages that may require validation.

Business Impact

Data quality issues can reduce confidence in analytical results and may lead to inaccurate business decisions.

Conclusion

Although the dataset was suitable for analysis, several quality issues should be addressed through improved data governance practices.

7. Relationship Between ASP and Unit Conversion

Objective

Determine whether Average Selling Price affects Unit Conversion for SKU C120[H:8NV.

Approach

Daily data for the SKU and a Pearson Correlation was analyzed and following metrics were calculated:

- Average Selling Price (ASP)
- Unit Conversion Rate

Findings

Correlation Result

Metric	Value
Correlation Coefficient	0.006345

Statistical Interpretation

- +1 indicates strong positive relationship
- -1 indicates strong negative relationship
- 0 indicates no meaningful relationship

The observed value of 0.006345 is extremely close to zero.

Business Insight

Price fluctuations did not significantly impact customer conversion behavior for this SKU.

Following factors may have played a greater role in influencing purchases:

- Product demand
- Customer reviews
- Product availability
- Brand preference

Conclusion

No meaningful relationship was found between ASP and Unit Conversion.

Key Recommendations

Based on the analysis, the following recommendations are proposed:

Recommendation 1

Investigate the Keyboards - DELETED subcategory to determine the cause of its significant decline.

Recommendation 2

Review products that stopped selling after July to identify inventory or demand issues.

Recommendation 3

Improve data quality controls to prevent zero-unit records and obsolete product classifications.

Recommendation 4

Continue monitoring promotional events and evaluate their long-term effectiveness.

Recommendation 5

Focus on customer experience, product availability, and marketing initiatives rather than relying solely on pricing strategies to improve conversion.

Final Conclusion

This case study successfully transformed raw sales and customer engagement data into actionable business insights.

The analysis identified key revenue drivers, promotional opportunities, underperforming product segments, and important data quality concerns. A significant sale event was detected, and its impact was evaluated. Additionally, statistical analysis demonstrated that pricing did not significantly influence customer conversion for the analyzed SKU.

Overall, the project highlights the value of structured data analysis in supporting business decision-making and demonstrates the practical application of Business Analyst techniques in a real-world scenario.

Prepared By: Abhijeet Kedar

Role: Aspiring Business Analyst

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